

HUMAN PHYSIOLOGY

Exploration of living phenomena through cellular, molecular, and systems biology perspectives. Physiological processes emerge from regulatory networks connecting cells, tissues, organs, and systems.

Chapters in Unit 5:

Ch 14: Respiration

Ch 16: Excretion

Ch 18: Neural Control

Ch 15: Body Fluids

Ch 17: Locomotion

Ch 19: Chemical Coord.



Alfonso Corti (1822 – 1888)

SCIENTIST PROFILE

Italian anatomist who began his scientific career studying the cardiovascular systems of reptiles before turning attention to the mammalian auditory system. In **1851**, he published a paper describing the structure on the basilar membrane of the cochlea containing hair cells that convert sound vibrations into nerve impulses — named the **Organ of Corti** in his honor.

CHAPTER 14 COMPLETE DETAILED NOTES

Breathing and Exchange of Gases

Full Textbook Preservation • Exam-Oriented • Process Diagrams • Capacity Formulas

NCERT Pages 183–192

Target: NEET / Board Exams

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14.4 Transport of Gases

14.5 & 14.6 Regulation & Disorders

CHAPTER MIND MAP — BREATHING & EXCHANGE OF GASES

1. Organs & Structure

- Simple Diffusion (Invertebrates)
- Cutaneous / Tracheal / Branchial
- Pulmonary (Human System)
- Conducting vs Exchange Part
- Thoracic Chamber Anatomy

2. Breathing Mechanism

- Pressure Gradients
- Inspiration (Negative Pressure)
- Expiration (Positive Pressure)
- Diaphragm & Intercostals
- Breathing Rate: 12-16/min

3. Volumes & Capacities

- Volumes: TV, IRV, ERV, RV
- Capacities: IC, EC, FRC, VC, TLC
- Clinical Spirometer Assessment

4. Gas Exchange & Transport

- Partial Pressures (pO_2 , pCO_2)
- 3-Layer Diffusion Membrane
- Oxyhaemoglobin & Sigmoid Curve
- CO_2 Bicarbonate & Carbonic Anhydrase

5. Regulation & Disorders

- Respiratory Rhythm Centre (Medulla)
- Pneumotaxic Centre (Pons)

- Chemosensitive Area & Receptors
- Asthma, Emphysema, Fibrosis

14.1 RESPIRATORY ORGANS & HUMAN RESPIRATORY SYSTEM

NCERT Pages 183–185

Biological Concept of Respiration / Breathing

Organisms utilize oxygen (O₂) to indirectly break down simple nutrient molecules like glucose, amino acids, and fatty acids to derive cellular energy for metabolic activities. Harmful carbon dioxide (CO₂) released during these catabolic reactions must be removed. **Breathing (Respiration)** is the process of continuous exchange of atmospheric O₂ with cellular CO₂.

Comparative Respiratory Mechanisms Across Animal Phyla

Animal Group / Organisms	Respiratory Mechanism / Structure	Type of Respiration	Habitat / Special Features
Lower Invertebrates (Sponges, Coelenterates, Flatworms)	Simple diffusion over entire body surface	Cutaneous / Surface Diffusion	Aquatic or moist environments; no specialized respiratory organs
Earthworms	Moist cuticle	Cutaneous Respiration	Moist skin facilitates direct diffusion into capillaries
Insects	Network of tracheal tubes	Tracheal Respiration	Air tubes transport atmospheric air directly throughout the body
Aquatic Arthropods & Molluscs	Specialized vascularised structures called Gills	Branchial Respiration	Extract dissolved oxygen from aquatic surroundings
Fishes	Gills	Branchial Respiration	Vertebrates adapted to aquatic respiration
Amphibians (e.g., Frogs)	Moist skin (Cutaneous) and Lungs (Pulmonary)	Dual (Cutaneous + Pulmonary)	Skin used in water/hibernation; lungs used on land

Animal Group / Organisms	Respiratory Mechanism / Structure	Type of Respiration	Habitat / Special Features
Reptiles, Birds, Mammals	Vascularised bags called Lungs	Pulmonary Respiration	Terrestrial vertebrates with highly developed respiratory systems

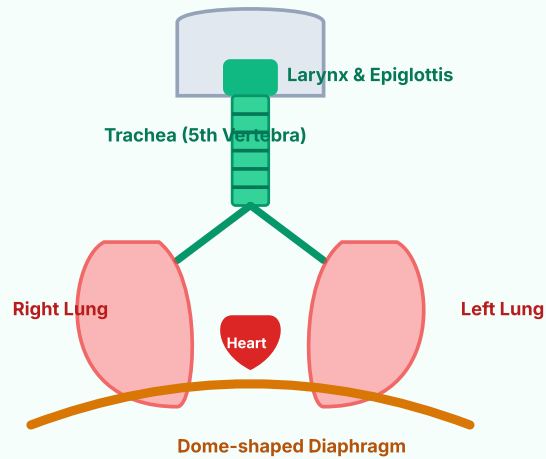
📌 **Easy Way to Remember Animal Respiratory Types:**

C-T-B-P: Cutaneous (Skin - Frog/Worm) → Tracheal (Tubes - Insects) → **B**ranchial (Gills - Fish/Molluscs) → **P**ulmonary (Lungs - Mammals/Birds/Reptiles).

Sequential Pathway of Human Respiratory Tract

1. **External Nostrils:** Pair of openings situated above the upper lips leading into the nasal passage.
2. **Nasal Chamber:** Passage lined with mucous membrane to filter, warm, and humidify incoming air.
3. **Pharynx:** Common chamber/passage for both food and atmospheric air.
4. **Larynx & Glottis:** Cartilaginous box assisting sound production (**Sound Box**). During swallowing, the glottis is covered by a thin elastic cartilaginous flap called the **Epiglottis** to prevent food entry.
5. **Trachea:** Straight tube extending to the mid-thoracic cavity. Divides at the level of the **5th thoracic vertebra** into right and left primary bronchi.
6. **Bronchial Tree:** Primary bronchi → Secondary bronchi → Tertiary bronchi → Initial bronchioles → Terminal bronchioles.
7. **Alveoli:** Thin, irregular-walled, highly vascularised bag-like structures arising from terminal bronchioles.

Figure 14.1 — Schematic View of Human Respiratory System



Cartilaginous Rings support: Trachea, primary, secondary, and tertiary bronchi, and initial bronchioles to prevent collapse.

1. Conducting Part

Extent: External nostrils up to terminal bronchioles.

Functions:

- Transports atmospheric air to the alveoli
- Clears air from foreign particles/dust
- Humidifies incoming air
- Brings air temperature to body temperature

2. Respiratory / Exchange Part

Extent: Alveoli and their ducts.

Function:

- Site of actual diffusion of O_2 and CO_2 between atmospheric air and blood
- Extremely thin wall for rapid gas diffusion

Double Pleural Membrane Protection

Lungs are enclosed by double-layered **Pleura** containing **pleural fluid** that reduces friction on the lung surface.

- **Outer Pleural Membrane:** In close contact with the thoracic lining.

- **Inner Pleural Membrane:** In direct contact with the lung surface.

Anatomically Air-Tight Thoracic Chamber

- **Dorsally:** Vertebral column
- **Ventrally:** Sternum
- **Laterally:** Ribs
- **Lower Side:** Dome-shaped Diaphragm

*Anatomy ensures any change in thoracic volume is directly reflected in pulmonary volume (essential for breathing as we cannot alter pulmonary volume directly).

5 FUNDAMENTAL STEPS INVOLVED IN RESPIRATION

Step 1

Breathing / Pulmonary Ventilation (Inflow of atmospheric O₂ & outflow of CO₂-rich alveolar air)

Step 2

Gas Diffusion across Alveolar Membrane

Step 3

Gas Transport by Blood

Step 4

Gas Diffusion between Blood & Tissues

Step 5

Cellular O₂ Utilisation & CO₂ Release

Pressure Gradient Rule of Breathing

Air movement into and out of lungs is driven by a **pressure gradient** between lungs and atmosphere. **Inspiration** occurs when intra-pulmonary pressure is lower than atmospheric pressure (negative pressure). **Expiration** occurs when intra-pulmonary pressure is higher than atmospheric pressure.

Detailed Mechanical Comparison: Inspiration vs Expiration

Parameter	Inspiration (Inhalation)	Expiration (Exhalation)
Definition	Process by which atmospheric air is drawn into the lungs.	Process by which alveolar air is expelled out of the lungs.
Pressure State	Intra-pulmonary pressure < Atmospheric pressure (Negative intra-pulmonary pressure).	Intra-pulmonary pressure > Atmospheric pressure (Positive intra-pulmonary pressure).
Diaphragm Action	Contracts and moves downwards (flattens), increasing thoracic volume in Antero-posterior axis .	Relaxes and arches upwards to dome shape, decreasing thoracic volume in antero-posterior axis.
Intercostal Muscles	External intercostal muscles contract , lifting ribs and sternum.	External intercostals relax; sternum and ribs return to original position.
Thoracic Axis Expansion	Increases thoracic volume in Dorso-ventral axis .	Reduces thoracic volume in dorso-ventral axis.
Pulmonary Volume & Pressure	Pulmonary volume increases → Intra-pulmonary pressure decreases below atmospheric → Air rushes IN.	Pulmonary volume decreases → Intra-pulmonary pressure increases above atmospheric → Air forced OUT.

★ Abdominal Muscles

Humans possess the ability to increase the strength of inspiration and expiration using additional muscles in the **abdomen**.

★ Normal Breathing Rate

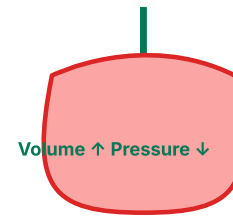
A healthy adult human breathes on average **12 – 16 times per minute** under physiological rest conditions.

★ Clinical Spirometer

The volume of air involved in breathing movements is measured using a **Spirometer** for clinical assessment of pulmonary functions.

Figure 14.2 — Visual Comparison of Breathing Mechanism

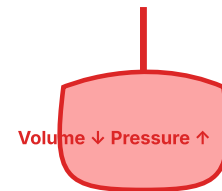
(a) INSPIRATION



Diaphragm Contracted (Flattened)

Ribs & Sternum raised • Thoracic cavity expands

(b) EXPIRATION



Diaphragm Relaxed (Arched Up)

Ribs & Sternum return • Air expelled out

Clinical Importance of Volumes & Capacities

Measurement of respiratory volumes and derivation of pulmonary capacities are essential diagnostic tools in clinical medicine for evaluating pulmonary health, lung compliance, and respiratory disorders.

1. Respiratory Volumes (Measured Directly by Spirometer)

Volume Name	Abbreviation	Average Volume Value	Complete NCERT Definition
Tidal Volume	TV	~500 mL (6000–8000 mL/min)	Volume of air inspired or expired during a normal respiration. A healthy man can inspire or expire approx 6000–8000 mL per minute.
Inspiratory Reserve Volume	IRV	2500 mL – 3000 mL	Additional volume of air a person can inspire by a forcible inspiration.
Expiratory Reserve Volume	ERV	1000 mL – 1100 mL	Additional volume of air a person can expire by a forcible expiration.
Residual Volume	RV	1100 mL – 1200 mL	Volume of air remaining in the lungs even after a forcible expiration. <i>(Cannot be measured directly by simple spirometer).</i>

2. Pulmonary Capacities (Derived by Combining Volumes)

Capacity Name	Abbreviation	Formula	Approximate Value	Complete NCERT Explanation
Inspiratory Capacity	IC	TV + IRV	~3000 – 3500 mL	Total volume of air a person can inspire after a normal expiration.
Expiratory Capacity	EC	TV + ERV	~1500 – 1600 mL	Total volume of air a person can expire after a normal inspiration.

Capacity Name	Abbreviation	Formula	Approximate Value	Complete NCERT Explanation
Functional Residual Capacity	FRC	ERV + RV	~2100 – 2300 mL	Volume of air that remains in lungs after a normal expiration.
Vital Capacity	VC	ERV + TV + IRV	~3500 – 4700 mL	Maximum volume of air a person can breathe in after a forced expiration (or breathe out after forced inspiration).
Total Lung Capacity	TLC	RV + ERV + TV + IRV (or VC + RV)	~4600 – 5800 mL	Total volume of air accommodated in lungs at the end of a forced inspiration.

🧠 Easy Memory Tricks for Capacities:

- **IC** = Tidal + Inspire (TV + IRV)
- **EC** = Tidal + Expire (TV + ERV)
- **FRC** = What stays after normal breath = **ERV + RV**
- **VC** = Total usable air = **ERV + TV + IRV**
- **TLC** = Absolute Maximum Air = **Vital Capacity + Residual Volume**

★ Practice Calculation Note (Hourly Tidal Volume):

If normal TV = 500 mL, and breathing rate = 12 to 16 times/min:

Per minute volume = 500×12 to $500 \times 16 = 6000$ to 8000 mL/min\$.

Hourly Volume: 6000×60 to $8000 \times 60 = \mathbf{360,000}$ mL (360 L) to $\mathbf{480,000}$ mL (480 L) per hour\$.

Primary Sites & Diffusion Factors

Alveoli are the primary sites of gas exchange. Exchange also occurs between blood and body tissues. Gas exchange occurs by simple diffusion based primarily on **partial pressure gradients, solubility of gases, and thickness of diffusion membrane.**

TABLE 14.1 — Partial Pressures (in mm Hg) of Oxygen and Carbon Dioxide

Respiratory Gas	Atmospheric Air	Alveoli	Deoxygenated Blood	Oxygenated Blood	Body Tissues
Oxygen (O ₂)	159	104	40	95	40
Carbon Dioxide (CO ₂)	0.3	40	45	40	45

*The table shows a steep concentration gradient for O₂ from alveoli (104) → blood (40) → tissues (40). Conversely, CO₂ gradient flows from tissues (45) → blood (45) → alveoli (40).

★ Critical Fact: Solubility Difference Between Gases

The solubility of CO₂ is **20 to 25 times higher** than that of O₂. Therefore, the amount of CO₂ that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O₂.

★ Critical Fact: Solubility Difference Between Gases

The solubility of \$CO_2\$ is **20 to 25 times higher** than that of \$O_2\$. Therefore, the amount of \$CO_2\$ that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of \$O_2\$.

The Three Major Layers of the Diffusion Membrane

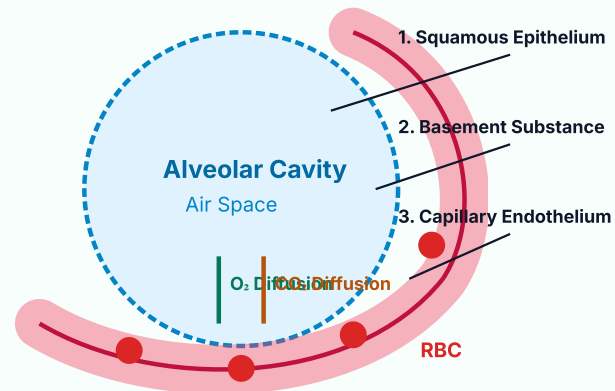
Total thickness of diffusion membrane is **much less than a millimeter (< 1 mm)**, facilitating rapid diffusion.

1. Thin Squamous Epithelium: Single-celled thick lining of the alveolar wall.

2. Endothelium of Alveolar Capillaries: Single-celled endothelial wall surrounding capillary blood vessels.

3. Basement Substance: Composed of a thin basement membrane supporting alveolar epithelium + basement membrane surrounding capillary endothelial cells.

Figure 14.4 — Diagram of Section of an Alveolus with Pulmonary Capillary



Total diffusion barrier thickness < 1 mm, creating ideal conditions for gas exchange.

Transport Mechanisms of Oxygen (O_2)

As Oxyhaemoglobin (in RBCs): **97%**

Dissolved State in Blood Plasma: **3%**

Transport Mechanisms of Carbon Dioxide (CO_2)

As Bicarbonate Ions (HCO_3^-): **70%**

As Carbamino-haemoglobin (RBCs): **20 – 25%**

Dissolved State in Plasma: **7%**

14.4.1 Transport of Oxygen & Oxygen Dissociation Curve

Haemoglobin is a red-coloured iron-containing pigment present in RBCs. O_2 binds reversibly with haemoglobin to form **Oxyhaemoglobin**. Each haemoglobin molecule can carry a maximum of **4 molecules of O_2** .

Oxygen Dissociation Curve Definition

A **Sigmoid curve** obtained when percentage saturation of haemoglobin with O_2 is plotted against partial pressure of oxygen (pO_2).

Conditions Favouring Binding in Alveoli (Shift to Left):

- High pO_2
- Low pCO_2
- Lesser H^+ ion concentration (higher pH)
- Lower temperature

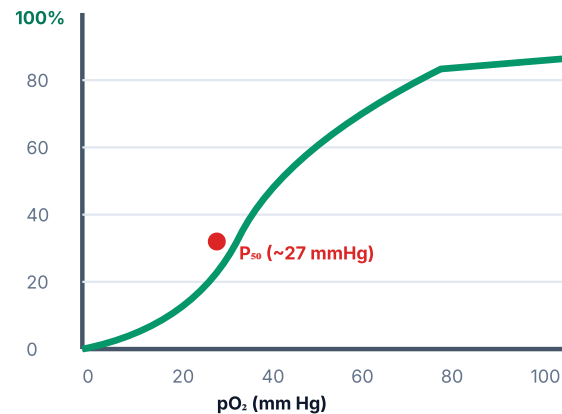
Conditions Favouring Dissociation in Tissues (Shift to Right):

- Low pO_2
- High pCO_2
- High H^+ ion concentration (lower pH)
- Higher temperature

★ NCERT Delivery Value:

Every **100 mL of oxygenated blood** delivers approximately **5 mL of O_2** to the tissues under normal physiological conditions.

Figure 14.5 — Oxygen Dissociation Curve (Sigmoid Graph)



Sigmoid shape reflects cooperative binding of oxygen to 4 haemoglobin subunits.

14.4.2 Transport of Carbon Dioxide & Carbonic Anhydrase Reaction

CO_2 is carried by haemoglobin as **Carbamino-haemoglobin** (20–25%). Binding is governed by pCO_2 . High pCO_2 and low pO_2 in tissues promote binding; low pCO_2 and high pO_2 in alveoli promote dissociation.

REVERSIBLE ENZYMATIC REACTION (CARBONIC ANHYDRASE)

RBCs contain an extremely high concentration of the enzyme **Carbonic Anhydrase** (minute quantities also exist in plasma).



At Tissue Site (High pCO_2): CO_2 diffuses into blood and forms HCO_3^- and H^+ .

At Alveolar Site (Low pCO_2): Reaction proceeds in reverse direction, converting HCO_3^- back to CO_2 and H_2O to be expired.

★ **NCERT Delivery Value for CO_2 :**

Every **100 mL of deoxygenated blood** delivers approximately **4 mL of CO_2** to the alveoli.

14.5 Neural & Chemical Control of Respiration

1. Respiratory Rhythm Centre

Located in the **Medulla oblongata** region of brain. Primary centre responsible for maintaining basic respiratory rhythm.

2. Pneumotaxic Centre

Located in the **Pons** region of brain. Moderates functions of rhythm centre. Signals **reduce duration of inspiration**, altering respiratory rate.

3. Chemosensitive Area

Situated adjacent to rhythm centre. Highly sensitive to **CO₂ and H⁺ ions**. Activates rhythm centre to speed up elimination when levels rise.

4. Aortic Arch & Carotid Receptors

Receptors associated with **aortic arch** and **carotid artery** recognize CO₂ and H⁺ concentration changes and send signals to rhythm centre.

★ Crucial NCERT Exception Rule:

The role of **Oxygen (O₂)** in the regulation of respiratory rhythm is **quite insignificant**. Regulation is primarily driven by CO₂ and H⁺ ion levels.

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14.6 Disorders of Respiratory System

1. Asthma

Difficulty in breathing causing **wheezing** due to **inflammation of bronchi and bronchioles**.

2. Emphysema

A **chronic disorder** in which alveolar walls are damaged, resulting in a permanent **decrease in respiratory surface area**. Major cause: **Cigarette smoking**.

3. Occupational Disorders

Seen in industries involving **grinding or stone-breaking**. Excessive dust causes inflammation leading to **Fibrosis** (proliferation of fibrous tissues) and serious lung damage. Protective masks are essential.



FINAL REVISION — QUICK RECALL SUMMARY

500 mL

Normal Tidal Volume (TV)

12 - 16 / min

Normal Breathing Rate

104 mm Hg

pO₂ in Alveoli

40 mm Hg

pCO₂ in Alveoli

5 mL O₂

Delivered per 100 mL blood

4 mL CO₂

Delivered per 100 mL blood

70% CO₂

Transported as Bicarbonate

97% O₂

Transported as Oxyhaemoglobin



NCERT CHAPTER EXERCISES & DETAILED SOLUTIONS

Page 192 • Questions 1 to 14

Q1. Define vital capacity. What is its significance?

Answer: Vital Capacity (VC) is defined as the maximum volume of air a person can breathe in after a forced expiration (or the maximum volume of air a person can breathe out after a forced inspiration). Formula: $VC = ERV + TV + IRV$.

Significance: It indicates the maximum working capacity and strength of the respiratory muscles and lungs. Higher vital capacity provides greater ability to supply O₂ during intense muscular exercise.

Q2. State the volume of air remaining in the lungs after a normal breathing.

Answer: The volume of air remaining in lungs after normal expiration is called **Functional Residual Capacity (FRC)**.

$FRC = ERV + RV = 1000\text{--}1100\text{ mL} + 1100\text{--}1200\text{ mL} = \mathbf{2100\text{ to }2300\text{ mL}}$.

Q3. Diffusion of gases occurs in the alveolar region only and not in the other parts of respiratory system. Why?

Answer: Diffusion occurs exclusively in alveoli because alveoli possess an extremely thin diffusion membrane (< 1 mm) made of single-celled squamous epithelium closely lined with blood capillaries, providing a huge surface area. Other conducting parts (trachea, bronchi) have thick cartilaginous rings and ciliated epithelium meant only for air transport.

Q4. What are the major transport mechanisms for CO_2 ? Explain.

Answer: CO_2 is transported by 3 mechanisms:

1. **As Bicarbonate Ions (70%):** CO_2 reacts with H_2O inside RBCs catalyzed by *Carbonic Anhydrase* to form H_2CO_3 , which dissociates into HCO_3^- and H^+ .
2. **As Carbamino-haemoglobin (20–25%):** CO_2 binds directly to the amino groups of haemoglobin in RBCs.
3. **Dissolved in Plasma (7%):** Transported in physical solution state in blood plasma.

Q5. What will be the pO_2 and pCO_2 in the atmospheric air compared to those in the alveolar air?

Answer: Correct Option: (ii) pO_2 higher, pCO_2 lesser.

Explanation: Atmospheric air has $\text{pO}_2 = 159$ mm Hg and $\text{pCO}_2 = 0.3$ mm Hg, whereas alveolar air has $\text{pO}_2 = 104$ mm Hg and $\text{pCO}_2 = 40$ mm Hg.

Q6. Explain the process of inspiration under normal conditions.

Answer: Inspiration is an active process initiated by: (1) Contraction of diaphragm flattening it and increasing thoracic volume in antero-posterior axis. (2) Contraction of external intercostal muscles lifting ribs and sternum, increasing volume in dorso-ventral axis. The overall increase in thoracic volume increases pulmonary volume, dropping intra-pulmonary pressure below atmospheric pressure, forcing atmospheric air into lungs.

Q7. How is respiration regulated?

Answer: Respiration is regulated neurally and chemically: (1) **Respiratory Rhythm Centre** in Medulla maintains basic rhythm. (2) **Pneumotaxic Centre** in Pons reduces inspiration duration. (3) **Chemosensitive Area** adjacent to rhythm centre responds to elevated CO_2 and H^+ ions. (4) Receptors in **Aortic arch & Carotid artery** detect CO_2 / H^+ changes and send corrective signals.

Q8. What is the effect of pCO_2 on oxygen transport?

Answer: High pCO_2 lowers haemoglobin affinity for O_2 (Bohr Effect), shifting oxygen dissociation curve to the right and promoting O_2 release to tissues. Conversely, low pCO_2 in alveoli promotes O_2 binding to haemoglobin.

Q9. What happens to the respiratory process in a man going up a hill?

Answer: At high altitudes, atmospheric pressure and pO_2 decrease, leading to hypoxia. To compensate, breathing rate increases, heart rate increases, and over time RBC production increases (stimulated by erythropoietin).

Q10. What is the site of gaseous exchange in an insect?

Answer: Tracheoles (the finest branches of the tracheal network) where direct gas exchange occurs with body tissues.

Q11. Define oxygen dissociation curve. Can you suggest any reason for its sigmoidal pattern?

Answer: It is the graphic plot of percentage saturation of haemoglobin against partial pressure of oxygen (pO_2).

Reason for Sigmoidal Pattern: Cooperative binding of oxygen. Binding of the first O_2 molecule to one haem subunit induces a conformational change that increases affinity of remaining subunits for O_2 .

Q12. Have you heard about hypoxia? Try to gather information about it, and discuss with your friends.

Answer: Hypoxia is a pathological condition where body tissues receive insufficient oxygen supply. Causes include high altitude (hypoxic hypoxia), anaemia (anaemic hypoxia), or toxic chemicals like carbon monoxide poisoning.

Q13. Distinguish between: (a) IRV and ERV (b) Inspiratory capacity and Expiratory capacity (c) Vital capacity and Total lung capacity.

Answer:

(a) **IRV vs ERV:** IRV is extra volume inspired forcibly (~2500–3000 mL); ERV is extra volume expired forcibly (~1000–1100 mL).

(b) **IC vs EC:** $IC = TV + IRV$ (~3000–3500 mL); $EC = TV + ERV$ (~1500–1600 mL).

(c) **VC vs TLC:** $VC = ERV + TV + IRV$ (~3500–4700 mL); $TLC = VC + RV$ (~4600–5800 mL).

Q14. What is Tidal volume? Find out the Tidal volume (approximate value) for a healthy human in an hour.

Answer: Tidal Volume (TV) is the volume of air inspired or expired during a normal breath (~500 mL).

Hourly Calculation:

Per minute volume = $500 \text{ mL} \times 12 \text{ to } 16 \text{ breaths/min} = 6000 \text{ to } 8000 \text{ mL/min}$.

Per hour volume = $6000 \text{ to } 8000 \text{ mL} \times 60 \text{ minutes} = \mathbf{360,000 \text{ mL (360 L) to } 480,000 \text{ mL (480 L)}}$.

